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Survival of endodontically treated permanent teeth among children: a retrospective cohort study

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Background

In the general population, multiple outcome studies have been performed regarding nonsurgical root canal treatment (NSRCT) with varying results. A number of factors influence this variation including the outcome measured, the study design, follow-up duration and operator skill and experience. The success rate of NSRCT ranges from 75 to 85%, depending on the strictness of the criteria

used [1]. The percentage of healed teeth ranges from 73 to 97%, while functional teeth range between 88 and 97% [2].

Few studies have investigated endodontic involvement and NSRCT in the permanent teeth of children. In a Turkish population, evaluation of panoramic radiographs and charts of 6- to 12-year-olds showed that 0.47% of permanent first molars were endodontically treated, while 4% required NSRCT [3]. In 13- to 16-year-olds, 4.28% of permanent first molars were endodontically treated and 6.09% required NSRCT [3]. Another study also examined patient charts in a Saudi population and

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reported that 35.8% of permanent teeth in 6- to 18-year-olds were pulpally involved [4].

Reports on the NSRCT outcomes in permanent teeth of this younger population are rare [5]. The technical quality of NSRCT was satisfactory in only 42–61% of cases, with a mean age of 12–16 years at treatment time [6–8]. Radiographic exams of endodontically treated teeth (ETT) showed healthy periapical tissue in 48–75% of teeth [6, 8]. Moreover, successful NSRCT was reported in 36–86% of teeth in 8- to 20-year-olds after clinical and radiographic exams [5, 9]. The marked difference in the previous studies' results could be attributed to differences in design and the criteria used to measure the outcome. The purpose of this study was to examine the survival rates of NSRCTs performed on permanent teeth of 6- to 18-year-olds at Boston University Henry M. Goldman School of Dental Medicine (BUGSDM). Predictors of failure of ETT were also assessed. Our null hypothesis was that age, gender, insurance type, tooth type and jaw type have no effect on treatment outcome (survival).

Methods

Study design, setting, and participants

In this retrospective analysis of electronic dental records of 6- to 18-year-olds, patients were selected after ethical approval was obtained from the Medical Center Institutional Review Board (IRB Number: H-34766). A total of 25,877 records were scanned to identify 773 patients (931 teeth) who received endodontic treatment between January 1st, 2007, and December 31st, 2015. Subjects with completed NSRCTs during the study period and at least one posttreatment record were included in the analysis. Hence, records of 432 patients (507 teeth) were excluded due to lack of follow-up, leaving a sample size of 341 patients with 424 teeth (Fig. 1). The patient data were deidentified and a study ID was used to record the findings. The NSRCT was mainly performed by endodontic residents who were trained to perform microscopic endodontic therapy using evidence-based protocols and techniques with specialist faculty supervision for uniform results.

Study outcome ascertainment

The main outcome variable was time until failure of ETT, i.e., time-to-event or time until a participant has the event of interest. If the ETT was symptomatic or subjected to any of the following interventions: extraction, endodontic retreatment, or apical surgery, it was deemed a failure. Otherwise, survival was defined as the absence of reported clinical signs or symptoms and the absence of a treatment intervention regardless of the radiographic interpretation.

Covariates

Data on the following covariates were extracted from the electronic dental records: age at time of treatment (6–11, 12–14, and 15–18 years), gender, insurance type (Medicaid [MassHealth] and private/self-paid), tooth type (posterior and anterior), and jaw type (maxillary and mandibular). Insurance type was used to reflect on the patient socioeconomic status (SES).

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^a Hazard ratios adjusted for all variables shown in the table

tors that may influence the outcome of RSKCTs, such as traumatic teeth with periapical radiolucency in the rupture

Conclusions

Age at treatment was a significant factor in determining the risk of ETT failure among children aged 6–18 years. The 5-year survival probability of ETT was 80.0% for 15- to 18-year-olds, 64.8% for 12- to 14-year-olds, and 46.4% for 6- to 11-year-olds. The average 5-year survival of ETT for all age groups was 69.1%. Our results suggest that the longer NSRCT is prevented, through proper oral hygiene measures, preventive dental care and less invasive procedures, the better the likelihood of survival of ETT in children. Vital pulp therapy should be considered over NSRCT when possible in younger children.

Abbreviations

NSRCT: Nonsurgical root canal treatment; ETT: Endodontically treated teeth; BUGSDM: Boston University Henry M. Goldman School of Dental Medicine; HR: Hazard ratio; CI: Confidence interval; KM: Kaplan–Meier.

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Authors' contributions

SB: Conceptualization, Investigation, Data curation, Writing—Original draft, Writing—Review and editing, Visualization. JJ: Conceptualization, Methodology, Writing—review and editing, Supervision, Project administration. JS: Writing—Review and editing, Supervision. TH: Writing—Review and editing. SR: Methodology, Software, Formal analysis, Data curation. AHZ: Writing—Methodology, Conducting statistical analysis and data interpretation, Original draft, Writing—Review and editing. SC: Conceptualization, Methodology, Writing—Review and editing, Supervision. All authors read and approved the final manuscript.

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Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Ethical approval was obtained from Boston University Medical Center Institutional Review Board (IRB Number: H-34766).

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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